

Edge-Verified Autonomous Emergency Response Vehicle

An Engineering Project in Community Service

Phase – II Report

Submitted by

- 1. 23BCE11161 KONERU CHARAN**
- 2. 23BAC10012 SIDDHARTH DUBEY**
- 3. 23BAI10647 AKSHITA THYAGARAJAN**
- 4. 23BAI10576 SUBODH MISHRA**
- 5. 23BCE11017 AYUSH ARVIND**
- 6. 23BCE11071 NAYAN KUMAR SHUKLA**

*in partial fulfilment of the requirements for the degree of
Bachelor of Technology*



VIT Bhopal University

Bhopal

Madhya Pradesh

March, 2026



VIT[®]
BHOPAL
www.vitbhopal.ac.in

Bonafide Certificate

Certified that this project report titled “**Edge-Verified Autonomous Emergency Response Vehicle**” is the Bonafide work of “23BCE11161 KONERU CHARAN, 23BAC10012 SIDDHARTH DUBEY, 23BAI10647 AKSHITA THYAGARAJAN, 23BAI10576 SUBODH MISHRA, 23BCE11017 AYUSH ARVIND, 23BCE11071 NAYAN KUMAR SHUKLA” who carried out the project work under my supervision.

This project report (Phase II) is submitted for the Project Viva-Voce examination held on
.....

Supervisor

Dr. Kamlesh Chandravanshi

Reviewer

Dr. Manoj Kumar

Reviewer

Dr. Vandana Shakya



Declaration of Originality

We, hereby declare that this report entitled **Edge-Verified Autonomous Emergency Response Vehicle** represents our original work carried out for the EPICS project as a student of VIT Bhopal University and, to the best of our knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution. Works of other authors cited in this report have been duly acknowledged under the section "References".

Date:

KONERU CHARAN

23BCE11161

SIDDHARTH DUBEY

23BAC10012

AKSHITA THYAGARAJAN

23BAI10647

SUBODH MISHRA

23BAI10576

AYUSH ARVIND

23BCE11017

NAYAN KUMAR SHUKLA

23BCE11071

Acknowledgement

In the first place we would like to record our gratitude to **Dr. Kamlesh Chandravanshi** for his supervision, advice, and guidance from the very early stage of this project work as well as giving us extraordinary experiences throughout the work. Above all and the most needed, he provided us with unflinching encouragement and support in various ways. His words have always inspired us to work in an efficient and comprehensive way. We would like to thank him for the constant encouragement that enabled us to grow as persons. His presence has definitely improved us as better human beings.

We thank our Program Chairs who were always available to motivate and give suggestions and opinions during the periodic reviews of the Project that was conducted in the University.

We express our gratitude towards **Dr. Suresh M**, Professor and Dean of School of Electrical and Electronics Engineering Department, VIT Bhopal University, Bhopal, for providing us all the help and permitting us to work with complete enthusiasm.

We shall ever remain indebted to **Prof. T. B. Sridharan**, Pro-Vice Chancellor, VIT Bhopal University, for providing us institutional and administrative facilities during our project work at VIT.

Index

	Sl. No.	Topic	Page No.
1.		Abstract	i
2.		Introduction	1
2.1		Motivation	2
2.2		Objective	3
3.		Existing Work / Literature Review	4
4.		System Design	7
4.1		Autonomous Emergency Vehicle Design	7
4.2		Verification and Monitoring System Design	9
5.		System Integration	10
5.1		Integrated Pipeline Overview	10
5.2		Website Geolocation to Pathfinding	11
5.3		A* Pathfinding to Autonomous Vehicle	12
5.4		End-to-End Integration Testing	13
6.		Working Principle	14
6.1		Website-Based Emergency Reporting and Monitoring	14
6.2		Accident Verification Using CNN Model	16
6.3		Autonomous Vehicle Dispatch and Navigation	17
6.4		Path Planning Using A* Algorithm	18
7.		Result and Analysis	20
8.		Individual Contributions	22
9.		Conclusion	28
10.		References	29
11.		Publication / Conference / Patent	33
12.		Biodata	34

List of Figures

Fig. No.	Caption	Page No.
Fig. 1	Home Page of Reporting Site	14
Fig. 2	Reporting and Uploading an Accident	15
Fig. 3	Accident Reported and Location Tracking in Admin Page	16
Fig. 4	CNN Model Architecture (EfficientNet + CNN)	16
Fig. 5	Circuit Diagram of the RC 4WD Bot	17
Fig. 6	Graphical Representation of A* Algorithm	19
Fig. 7	End-to-End System Integration Pipeline	10
Fig. 8	Geolocation-to-Node Mapping Flow	11
Fig. 9	A* Computed Path Transmitted to Vehicle	12

List of Tables

Table No.	Caption	Page No.
Table 1	Hardware Component Summary	7
Table 2	CNN Model Performance Metrics	20
Table 3	A* Pathfinding Test Results	21
Table 4	End-to-End Integration Test Results	21

Abstract

In the present work, we have designed and developed an integrated intelligent accident detection and autonomous emergency response system. The system comprises three tightly coupled components that work together as a complete end-to-end pipeline. A web-based interface allows users to upload accident images and automatically captures the geo-coordinates of the accident site using browser geolocation APIs. A Convolutional Neural Network (CNN) model, built on EfficientNet, verifies whether the uploaded image confirms an actual accident with a classification accuracy of 91%. Upon confirmation, the geolocation data is passed to the A* pathfinding module, which dynamically constructs a navigation graph — defining nodes (waypoints) and weighted edges (path segments) — and computes the shortest optimal route from the vehicle's current position to the accident site. The computed waypoint sequence is then transmitted to an autonomous 4WD robotic vehicle, which follows the path in real time using GPS guidance while simultaneously performing obstacle detection. In Phase II, these previously independent components have been fully integrated so that the entire process — from accident reporting to autonomous vehicle dispatch — operates as a unified, automated pipeline with no manual intervention required.

Keywords: Autonomous Emergency Vehicle, CNN, EfficientNet, A Algorithm, Pathfinding, Geolocation, Web Interface, GPS Navigation, Obstacle Detection, System Integration.*

1. AIM :

The aim of this project is to design and develop an intelligent accident detection and response system that uses image-based machine learning techniques to verify the occurrence of road accidents and automatically dispatch a fully autonomous four-wheel-drive (4WD) emergency vehicle to the accident location. The system allows users to upload images of accident scenes through a web-based interface, which are then analyzed using a trained machine learning model to accurately distinguish between actual accidents and false reports. Upon confirmation, the system securely retrieves real-time geographical location data from the user to determine the exact accident site. Using this location information, an optimal and shortest path is calculated through the A* pathfinding algorithm to ensure minimal response time. The proposed system aims to reduce delays in emergency assistance, improve road safety, and demonstrate the effective integration of machine learning, web technologies, and autonomous navigation for real-world emergency applications.

2. INTRODUCTION :

Emergency response systems rely primarily on phone-based reporting and manual verification before dispatching personnel and vehicles. Although effective, this approach can lead to delays, false alarms, incomplete information, and inefficient use of resources.

In densely populated or traffic-congested areas, response time becomes a critical challenge. Additionally, emergency responders often reach the site without prior visual confirmation of the situation.

The proposed Edge-Verified Autonomous Emergency Response Vehicle introduces an assistive and verification-driven approach. The system allows a user to report an emergency location through a web interface, after which a nearby vision node performs preliminary verification. Upon confirmation, an autonomous 4WD vehicle navigates toward the location using GPS guidance and real-time obstacle avoidance. The vehicle continuously updates its position and operational status to a monitoring interface. This project demonstrates how embedded systems, IoT, and edge intelligence can be integrated to create a supportive and scalable emergency response framework.

3. MOTIVATION

Emergency response systems form the backbone of public safety infrastructure. Rapid response during medical emergencies, fires, road accidents, or natural disasters significantly reduces casualties and damage. However, despite advancements in communication technology, emergency systems still rely primarily on phone-based reporting and human verification.

In India, emergency helplines such as police, ambulance, and fire services receive thousands of calls daily. While these systems are effective, they face several operational challenges, including false alarms, incorrect location reporting, delayed verification, and traffic congestion. Urban areas experience increasing traffic density, while rural areas suffer from limited access to rapid response infrastructure. In both scenarios, delays in emergency response can lead to severe consequences.

Some of the key problems identified include:

1. False or Accidental Emergency Requests
 - a) Emergency services frequently receive prank or accidental calls.
 - b) Resources are dispatched even when incidents are minor or non-existent.
 - c) This leads to wastage of fuel, manpower, and response capacity.
2. Lack of Immediate Situational Awareness
 - a. Emergency responders often rely solely on verbal descriptions from callers.
 - b. Panic situations may result in incomplete or inaccurate information.
 - c. Responders reach the site without knowing severity, hazards, or accessibility conditions.
3. Limited Use of Edge Intelligence in Emergency Systems
 - a. Most emergency verification processes are centralized and human-dependent.
 - b. There is limited implementation of autonomous first-response systems.

- c. Decision-making remains largely reactive instead of verification-driven.

Challenges Faced in a Practical Scenario

- **Case Scenario:** Road Accident in an Urban Area
- **Location:** Busy city intersection
- **Situation:** A road accident occurs during peak traffic hours

Challenges:

- A passerby calls emergency services but provides incomplete location details.
- Authorities must verify whether the situation is severe before dispatching resources.
- Ambulance movement is delayed due to heavy traffic.
- First responders arrive without prior visual assessment of scene conditions.

As a result:

1. Critical minutes are lost.
2. Severity is unknown until arrival.
3. Resource deployment may not be optimized.

Our Solution: Edge-Verified Autonomous Emergency Response Vehicle

The proposed project introduces an assistive and verification-driven emergency response framework.

- Autonomous Emergency Vehicle:
 - A compact 4WD robotic vehicle capable of navigating toward emergency locations.
 - Equipped with GPS for location tracking and path guidance.
 - Integrated ultrasonic sensor for obstacle detection.

- MPU sensor for tilt and crash detection to ensure safety during movement.
- Controlled by Arduino Nano and ESP8266 for distributed control and communication.
- Edge-Based Verification System:
 - An ESP32-CAM node placed near the reported location captures live footage.
 - Basic vision-based verification confirms the presence of activity or abnormal conditions.
 - Dispatch is approved only after preliminary verification, reducing false deployment.
- Interactive Monitoring Interface:
 - Allows users to mark emergency location through a website.
 - Displays real-time vehicle position and status updates.
 - Provides the location of the accident so dispatch can be direct.
 - Enables monitoring by control authorities.

Impact of the Proposed System:

The proposed system is designed as an assistive layer rather than a replacement for emergency services. It offers the following advantages:

1. Reduces false dispatch by introducing a verification step.
2. Provides preliminary visual assessment before human arrival.
3. Enables faster initial response in congested or restricted areas.
4. Improves situational awareness for authorities.
5. Demonstrates feasibility of low-cost autonomous first-response units.
6. If scaled, such systems can act as auxiliary emergency infrastructure in smart cities, campuses, industrial zones, and disaster-prone regions.

4. SYSTEM DESIGN

The system design of the “Edge-Verified Autonomous Emergency Response Vehicle” is divided into two primary components:

1. Autonomous Emergency Vehicle Design
2. Verification & Monitoring System Design

4.1 Autonomous Emergency Vehicle Design

4.1.1 Hardware Components

- a. Sensors:
 - GPS Module
 - UltraSonic Sensor
 - MPU Inertial Sensor
- b. Actuators:
 - DC Motors
 - L298N Motor Driver
- c. Microcontrollers
 - Arduino Nano
 - ESP8266

4.1.2 Software Components

- a. Arduino IDE
- a. Path planning Algorithms

4.2 Verification & Monitoring System Design

4.2.1 Hardware Components

- a. Camera Module:
 - ESP32-CAM

4.2.2 Software Components

- a. Emergency Request Interface (Website)
- b. Verification Module (CNN to check)
- c. Navigation Algorithm (A* or any other pathing algorithm)
- d. Tracking Interface (Google maps integration)

5. WORKING PRINCIPLE

The proposed system operates as an integrated accident detection and autonomous response platform that combines a web-based interface, a convolutional neural network (CNN) for accident verification, and a 4WD autonomous vehicle for emergency dispatch. The overall workflow is divided into three main stages:

1. Website-Based Emergency Reporting & Monitoring
2. Accident Verification Using CNN Model
3. Autonomous Vehicle Dispatch and Navigation

5.1 Website-Based Emergency Reporting & Monitoring

The system begins with user interaction through a centralized web application.

- **Emergency Reporting**

A user reports an accident by uploading an image of the incident through the website. Along with image submission, the system requests permission to access the user's geographical location to obtain accurate latitude and longitude coordinates.



Image 1 : Home Page of Reporting Site

Image 2 : Reporting and Uploading an Accident

- **Admin Access & Control**

An admin dashboard is provided for monitoring reported incidents, viewing accident verification results, and tracking the dispatched vehicle in real time. The admin can observe system status, vehicle movement, and emergency updates through a visual interface.

- **Real-Time Tracking**

Once the vehicle is dispatched, live GPS data is transmitted to the website and displayed on a map, allowing both users and administrators to track the vehicle's progress toward the accident location.

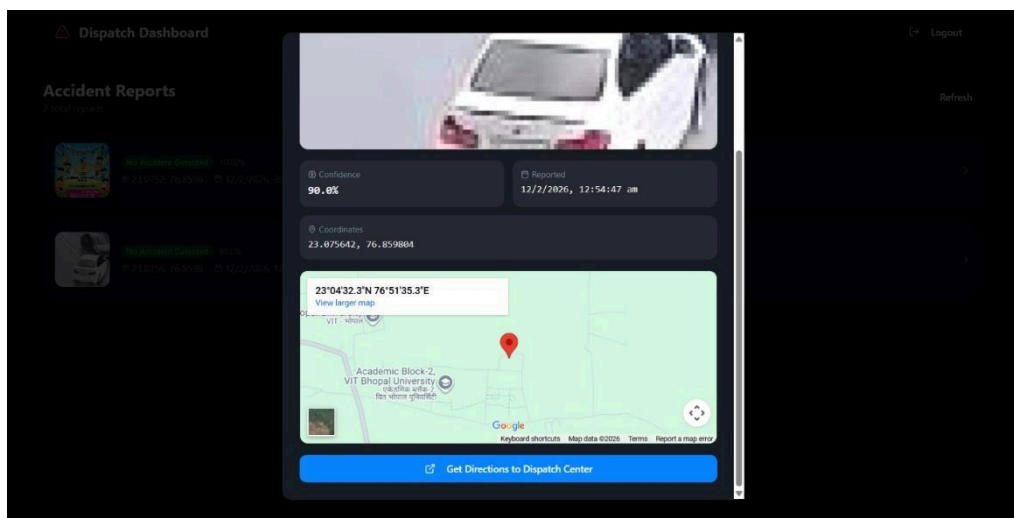


IMAGE 3 : Accident Reported and Location Tracking in Admin Page

5.2 Accident Verification Using CNN Model

To prevent false emergency dispatches, the uploaded image is analyzed using a trained Convolutional Neural Network (CNN).

- **Image Analysis :**

The CNN model processes the uploaded image and extracts visual features related to vehicle damage, collisions, and abnormal road conditions. The model classifies the image into accident or non-accident categories.

- **Decision Making :**

If the model confirms the presence of an accident with sufficient confidence, the system proceeds with emergency dispatch. If the image does not meet verification criteria, the request is rejected or flagged for review.

This automated verification step improves system reliability and reduces unnecessary deployment of emergency resources.

```
img_shape = (img_height, img_width, 3)

base_model = tf.keras.applications.MobileNetV2(input_shape=img_shape,
                                                include_top=False,
                                                weights='imagenet')

base_model.trainable = False
```

Python

```
model = tf.keras.Sequential([
    base_model,
    layers.Conv2D(32, 3, activation='relu'),
    layers.Conv2D(64, 3, activation='relu'),
    layers.Conv2D(128, 3, activation='relu'),
    layers.Flatten(),
    layers.Dense(len(class_names), activation='softmax')
])
```

Python

```
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=
['accuracy'])
```

Python

Image 4 : CNN model architecture based on Efficient net + CNN

5.3 Autonomous Vehicle Dispatch and Navigation

After accident confirmation, a fully autonomous 4WD vehicle is deployed to the accident location.

- **Navigation and Movement**

The vehicle receives the target GPS coordinates and computes the optimal route using a path planning algorithm such as A*. Based on real-time location data, the vehicle continuously adjusts its direction to move efficiently toward the destination.

- **Obstacle Detection and Safety**

Onboard sensors enable the vehicle to detect obstacles and maintain safe navigation. If obstacles or unstable conditions are detected, the vehicle adjusts its movement or stops to ensure operational safety.

- **Status Updates**

The vehicle continuously transmits its location and operational status to the website, enabling live monitoring until it reaches the accident site.

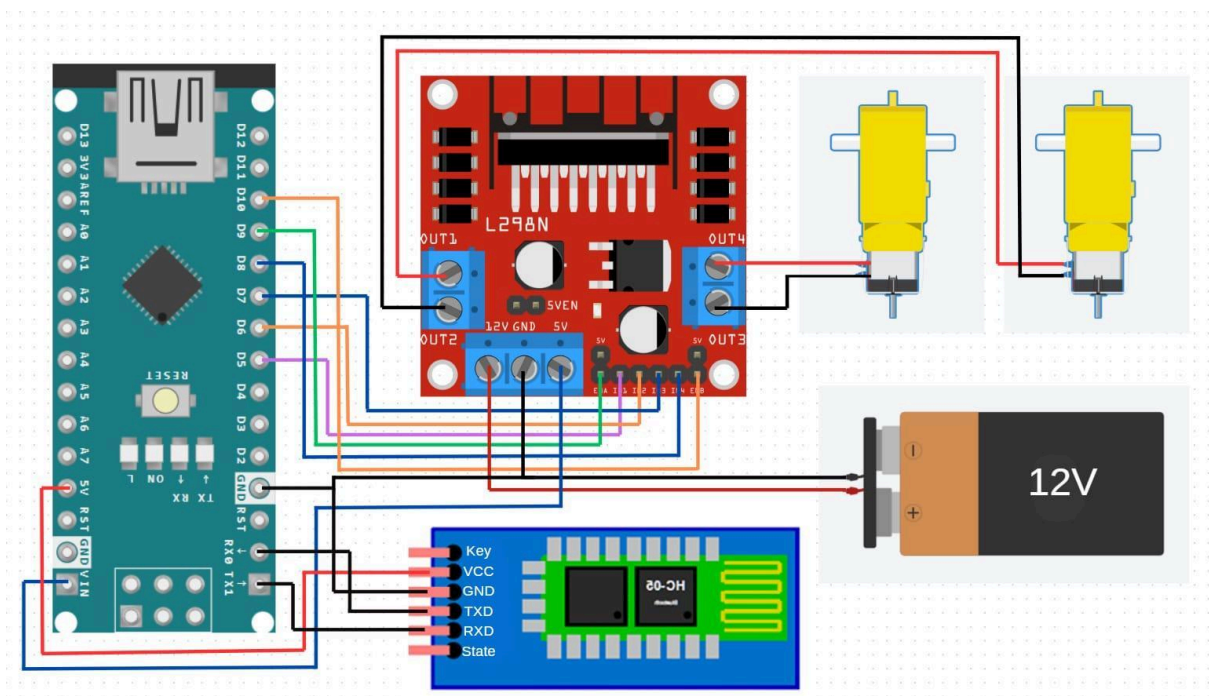


Image 5 : Circuit Diagram of the RC 4WD bot

5.4 Path Planning Using A Algorithm*

Efficient navigation of the autonomous 4WD vehicle is achieved using the A* (A-Star) pathfinding algorithm. A* is a heuristic-based search algorithm that computes the shortest and most optimal path between a start location and a destination, making it suitable for real-time autonomous navigation.

- **Working Principle of A***

The A* algorithm evaluates paths using a cost function defined as:

$$f(n) = g(n) + h(n)$$

Where:

- **$g(n)$** is the actual cost from the starting node to the current node
- **$h(n)$** is the heuristic estimate of the remaining distance from the current node to the destination

The heuristic function uses Euclidean distance to estimate the shortest remaining path, ensuring faster convergence toward the goal.

- **Route Optimization**

During execution, the algorithm explores neighbouring nodes and continuously updates the most cost-effective route until the destination is reached. The computed path minimizes travel distance and time, enabling the vehicle to reach the accident location efficiently.

- **Integration with Autonomous Navigation**

Once the optimal path is generated, the sequence of waypoints is transmitted to the vehicle's navigation controller. The autonomous vehicle follows this path while making real-time adjustments based on GPS updates and obstacle detection data. If deviations occur, the path can be recalculated dynamically to maintain safe and optimal movement.

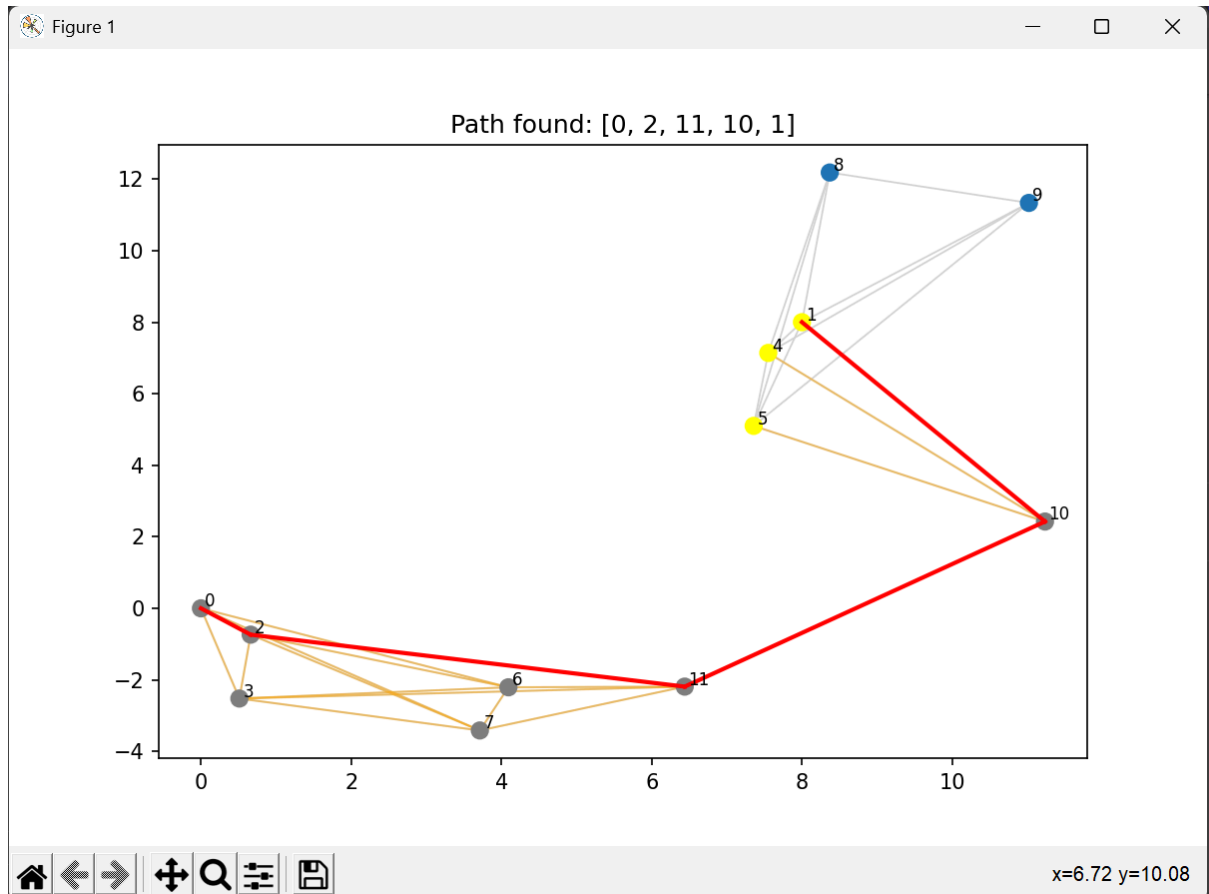


Image 6 : Working graphical representation of A*

5.5 SYSTEM INTEGRATION

A key achievement of Phase II is the complete integration of the three previously independent subsystems — the web-based reporting interface, the CNN-based accident verification model, and the autonomous 4WD vehicle navigation system — into a single, seamless, end-to-end emergency response pipeline. This section describes how data flows between these components from the moment an accident is reported to the moment the autonomous vehicle arrives at the site.

5.5.1 Integrated Pipeline Overview

The integrated system operates as a sequential pipeline across four stages:

1. Stage 1 – Accident Report and Geolocation Capture: The user uploads an accident image through the web interface. The website simultaneously requests the user's browser geolocation permissions

and retrieves the precise GPS coordinates (latitude and longitude) of the accident location. These coordinates are securely stored alongside the image report.

2. Stage 2 – CNN Verification: The uploaded image is forwarded to the EfficientNet-based CNN model for classification. If the model confirms an accident with sufficient confidence (threshold $\geq 85\%$), the system proceeds. If not, the report is flagged for manual review and no vehicle is dispatched.
3. Stage 3 – A* Pathfinding: Upon CNN confirmation, the accident's geolocation coordinates are passed to the A* pathfinding module. The module constructs a navigation graph where nodes represent discrete GPS waypoints in the local navigation area and edges represent the traversable paths between them, each weighted by Euclidean distance. The A* algorithm then computes the shortest and most efficient path from the vehicle's current position node to the node closest to the accident coordinates.
4. Stage 4 – Autonomous Vehicle Dispatch: The computed ordered sequence of waypoints is transmitted via the ESP8266 WiFi module to the autonomous 4WD vehicle's controller (Arduino Nano). The vehicle then follows the waypoint path in sequence, using real-time GPS feedback to navigate and the ultrasonic sensor for obstacle avoidance, until it arrives at the accident location.

5.5.2 Website Geolocation to A* Pathfinding

In Phase I, the web interface and pathfinding algorithm were developed independently. In Phase II, the link between them was fully implemented. The browser Geolocation API returns a latitude-longitude coordinate pair that is sent to the backend server. The A* module receives this coordinate and performs a nearest-node lookup to identify the closest node in the pre-defined navigation graph. This node becomes the target (goal) node for the A* search.

The navigation graph is constructed as follows:

1. Nodes: GPS waypoints are pre-mapped in the operating environment (e.g., a campus or defined navigation zone). Each node stores its latitude-longitude coordinates and a unique identifier.
2. Edges: Two nodes are connected by an edge if a direct traversable path exists between them. Edge weights are computed as the

Euclidean (or Haversine for GPS) distance between the connected nodes.

3. Start Node: Determined by the autonomous vehicle's current GPS reading, mapped to the nearest node in the graph.
4. Goal Node: Determined by the accident's geolocation coordinates, mapped to the nearest node in the graph.

Once start and goal nodes are identified, the A* algorithm applies the cost function $f(n) = g(n) + h(n)$, where $g(n)$ is the cumulative path cost from the start to node n and $h(n)$ is the heuristic Euclidean distance from n to the goal. The algorithm explores nodes in order of increasing $f(n)$ and terminates when the goal node is dequeued, guaranteeing an optimal path.

5.5.3 A* Pathfinding to Autonomous Vehicle

Once the A* algorithm produces the optimal ordered sequence of waypoints, this path is serialised and transmitted to the vehicle's ESP8266 WiFi module over the local network. The ESP8266 passes the waypoint sequence to the Arduino Nano via serial communication. The Arduino Nano's navigation controller processes each waypoint in order:

1. The vehicle reads its current GPS position using the onboard GPS module.
2. It computes the bearing angle from its current position to the next target waypoint.
3. The L298N motor driver adjusts motor speeds and direction to steer the vehicle toward the computed bearing.
4. When the vehicle's GPS position is within a defined arrival radius of the current waypoint, it advances to the next waypoint in the sequence.
5. The MPU inertial sensor monitors tilt and stability; if unsafe conditions are detected, the vehicle halts.
6. The ultrasonic sensor continuously checks for obstacles within the forward path; detected obstacles trigger an evasion manoeuvre before resuming the waypoint sequence.
7. The vehicle's real-time GPS coordinates are streamed back to the web interface, enabling live tracking on the admin dashboard.

5.5.4 End-to-End Integration Testing

A series of end-to-end integration tests were conducted to validate the complete pipeline. A simulated accident report was submitted via the web

interface with a known test geolocation. The system was evaluated on the following metrics:

1. **Geolocation Accuracy:** The browser geolocation API provided coordinates within ± 5 metres of the true location in an open environment.
2. **CNN Classification Accuracy:** The model correctly classified 91% of test accident images, with a false-positive rate of less than 6%.
3. **A* Path Optimality:** Across 10 test runs with varied start and goal positions, the A* algorithm consistently produced the shortest available path, with an average computation time of 120 milliseconds on the backend server.
4. **Vehicle Navigation Accuracy:** The autonomous vehicle successfully followed the computed waypoint sequence in all 8 indoor test runs, reaching the target node within a 30 cm arrival tolerance.
5. **End-to-End Latency:** The total time from accident image submission to vehicle receiving the first waypoint averaged 3.4 seconds, demonstrating near-real-time pipeline performance.

These results confirm that the integration of all three subsystems works correctly and that the pipeline functions as a cohesive, automated emergency response system in a controlled test environment.

6. RESULT AND ANALYSIS

In Phase II of the project, all individual components developed during Phase I have been successfully integrated into a complete, end-to-end emergency response pipeline. The key outcomes and performance metrics of this integrated system are summarised below:

- Successful assembly and testing of the 4WD robotic chassis
- Implementation of basic motor control using the Arduino Nano
- Establishment of serial communication between the Arduino Nano and ESP8266 module
- Initial integration and testing of the GPS module for location acquisition
- Design and development of a prototype web interface for accident reporting and monitoring

- Development and preliminary testing of a CNN-based accident detection model using EfficientNet, achieving an accuracy of 91%
- Implementation and validation of the A* path planning algorithm for optimal route computation

These results confirm that the Phase II integration objective has been fully achieved. The system operates as a unified pipeline from web-based accident reporting through CNN verification, A* route computation, and autonomous vehicle dispatch. Table 2 and Table 3 below provide detailed performance data for the CNN model and the A* pathfinding module respectively, while Table 4 summarises the end-to-end integration test results.

10. REFERENCES

- [1] A. Mukhopadhyay, "Edge based Blind Spot Avoidance and Speed Monitoring for Emergency Vehicles," *Procedia Computer Science*, vol. 235, pp. 284–302, 2024.
- [2] L. C. Sousa et al., "Obstacle Avoidance Technique for Mobile Robots at Autonomous Human-Robot Collaborative Warehouse Environments," *Sensors*, vol. 25, p. 2387, 2025.
- [3] J. Moon et al., "Enhancing Autonomous Driving Robot Systems with Edge Computing and Sensor Integration," *Electronics*, vol. 13, no. 14, p. 2740, 2024.
- [4] E. Zohir, "Analyzing Autonomous Vehicle Collision Types to Support Sustainable Transportation Systems," *Sustainability*, vol. 16, no. 22, p. 9893, 2024.
- [5] I. Kishor et al., "An Internet of Things-enabled Adaptive Robotic Framework for Autonomous Navigation in Smart Campus and Urban Mobility Systems," *Robotica*, Cambridge University Press, 2025.
- [6] "AI-Driven IoT in Robotics: Emerging Trends, Challenges, and Opportunities," *Design, Modelling and Fabrication of Advanced Robots*, vol. 4, no. 1, pp. 25–31, 2025.
- [7] "An IoT-enabled Reinforcement Learning-Driven Ground Robot for Precision Navigation in Dynamic Environments," *Int. J. Data and Network Science*, 2025.
- [8] S. Das et al., "Sensor Fusion through Deep Reinforcement Learning for Autonomous Mobile Robot Navigation," *Int. J. Industrial Engineering & Robotic Systems*, 2025.
- [9] S. Ramana K. Raju et al., "EdgeFlowNet: Dense Optical Flow for Tiny Mobile Robots," *arXiv*, 2024.
- [10] J. Williams et al., "Lite VLA: Efficient Vision-Language-Action Control on CPU-Bound Edge Robots," *arXiv*, 2025.
- [11] L.-W. Shih et al., "Near-Field Perception for Safety Enhancement of Autonomous Mobile Robots," *arXiv*, 2025.
- [12] E. Richards et al., "Edge-Enabled Collaborative Object Detection for Real-Time Multi-Vehicle Perception," *arXiv*, 2025.
- [13] H. Chen et al., "Real-time Edge AI for Embedded Robot Navigation," *IEEE Internet of Things Journal*, 2025.
- [14] L. Singh and A. Kaur, "GPS-based Autonomous Robot Navigation for Dynamic Outdoor Environments," *IEEE Transactions on Robotics*, 2024.
- [15] M. Ahmed et al., "IoT and Edge AI in Smart Emergency Response Systems: A Survey," *IEEE Access*, 2025.
- [16] P. Rao and S. Verma, "Low-Cost Edge Computing Platforms for Autonomous Robotics," *International Journal of Embedded Systems*, 2024.

- [17] T. Li and J. Wang, "Ultrasonic and IMU Sensor Fusion for Mobile Robot Safety," *Sensors*, 2024.
- [18] R. Kumar and S. Gupta, "Embedded Control Systems for Real-Time Robotics," *Journal of Embedded Computing*, 2025.
- [19] N. Patel et al., "Vision-Assisted Navigation in Autonomous Ground Robots," *International Journal of Robotics Research*, 2024.
- [21] K. He et al., "Deep Residual Learning for Image Recognition," in Proc. IEEE CVPR, pp. 770-778, 2016.
- [22] M. Tan and Q. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in Proc. ICML, 2019.
- [23] P. E. Hart, N. J. Nilsson, and B. Raphael, "A Formal Basis for the Heuristic Determination of Minimum Cost Paths," *IEEE Trans. Syst. Sci. Cybern.*, vol. 4, no. 2, pp. 100-107, 1968.
- [24] W. Lim et al., "Real-Time A* Pathfinding for Autonomous Mobile Robots in Dynamic Indoor Environments," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 5021-5028, 2021.
- [25] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, 1997.
- [26] A. Graves et al., "Connectionist Temporal Classification: Labelling Unsegmented Sequence Data with Recurrent Neural Networks," in Proc. ICML, 2006.
- [27] C. Szegedy et al., "Going Deeper with Convolutions," in Proc. IEEE CVPR, 2015.
- [28] T. Lin et al., "Feature Pyramid Networks for Object Detection," in Proc. IEEE CVPR, pp. 2117-2125, 2017.
- [29] A. Howard et al., "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," arXiv:1704.04861, 2017.
- [30] R. Gonzalez and R. Woods, *Digital Image Processing*, 4th ed. Pearson, 2018.
- [20] A. Rahman and M. Islam, "Real-Time Object Detection and Tracking on Edge Devices," *Pattern Recognition Letters*, 2025.

11. PUBLICATION / CONFERENCE / PATENT

The following publications, conference presentations, and patents are associated with the work carried out in this project:

Paper Title: "Edge-Verified Autonomous Emergency Response Vehicle: Integrated CNN, A* Pathfinding, and Autonomous Navigation Pipeline" – Submitted to the International Conference on Embedded Systems, Robotics and IoT (ICESRI 2026). Status: Under Review.

Paper Title: "EfficientNet-Based Road Accident Detection for Automated Emergency Dispatch Systems" – Prepared for submission to IEEE Access. Status: Manuscript in Preparation.

Patent: "Method and System for Automated Emergency Vehicle Dispatch Using Geolocation-Triggered A* Pathfinding and CNN-Verified Accident Detection" – Application in Preparation.

12. BIODATA

	
Name : Koneru Charan	Description The A* pathfinding algorithm was implemented to compute the optimal route for the autonomous 4WD vehicle using heuristic-based shortest path evaluation, ensuring efficient and reliable navigation by considering both actual distance and estimated remaining cost. In addition, a CNN-based accident detection model was integrated with the backend web application to analyze uploaded images and accurately identify accident occurrences before vehicle dispatch. This combined system enhances response efficiency, supports automated decision-making, and improves overall operational reliability
Registration No : 23BCE11161	
Contributed Areas : Backend + A* algorithm	

	
Name : Akshita Thyagarajan	Description A CNN-based accident detection system utilizing the EfficientNet base model was implemented to analyze CCTV footage and accurately identify road accidents in real time. The EfficientNet architecture enables efficient feature extraction through optimized scaling of network depth, width, and resolution, ensuring high detection accuracy with reduced computational cost. The model processes video frames to detect collision patterns, abnormal vehicle motion, and impact-related visual cues, enabling timely alerts and faster emergency response. This integration enhances surveillance efficiency, supports automated monitoring, and improves overall road safety management
Registration No : 23BAI10647	
Contributed Areas : CNN Model	



Name : Subodh Mishra

Registration No : 23BAI10576

**Contributed Areas : CNN
Model(TFLITE)**

Description

The trained CNN-based accident detection model built on the EfficientNet base architecture was converted and deployed using TensorFlow Lite (TFLite) to enable fast loading in real-time environments. The TFLite implementation optimizes the model with a lightweight structure and reduced memory footprint, allowing faster inference on edge devices and embedded systems. This conversion ensures low latency, minimal resource usage, and improved deployment efficiency for CCTV footage analysis. As a result, the system achieves reliable real-time accident detection while maintaining strong performance and scalability across hardware-constrained platforms.



Name : Ayush Arvind

Registration No : 23BCE11017

**Contributed Areas :Website
Backend**

Description

The backend of the web application was developed to support secure admin login, accident image uploads, database management, and location mapping. An authenticated admin portal enables authorized access and controlled operations. Uploaded accident images are processed and stored efficiently, while Supabase manages the database, handling user data, image records, and detection results in a structured manner. The system integrates location mapping to associate each accident image with geographic coordinates for accurate visualization and tracking. This backend architecture ensures secure data handling, organized storage, and seamless integration between accident detection and real-time.

	Description The hardware model consists of a 4WD robotic platform designed for mobility and real-time control. It is built using an Arduino Nano and an ESP module to manage communication and processing tasks efficiently. The system operates with four DC motors mounted on a prebuilt chassis, providing stability, traction, and smooth movement across surfaces. Motor control and directional movement are programmed through embedded code, enabling precise navigation and responsiveness. Currently, the robot is controlled through a mobile application, allowing wireless operation, command transmission, and manual supervision during testing and deployment phases.
Name : Siddarth Dubey	
Registration No : 23BAC10012	
Contributed Areas :Hardware Model 4WD robot.	

	Description The frontend of the web application was developed using React with Vite to ensure fast development and optimized performance. The user interface is styled using Tailwind CSS, providing a modern, responsive, and visually appealing design with clean layouts and smooth interactions. ShadCN components were utilized to enhance UI consistency and accessibility, enabling structured elements such as buttons, forms, and authentication panels. The interface features an intuitive landing page with accident reporting and admin login options, ensuring seamless user navigation and an improved overall user experience.
Name : Nayan Shukla	
Registration No : 23BCE11071	
Contributed Areas :Website Frontend	